

CARBON AND ITS COMPOUNDS

Student Learning Pack

1. Metadata

Topic	Carbon and its Compounds
Level	Class X
Lesson Length	60 minutes

2. Learning Objectives

- LO1: Explain how carbon's tetravalency and catenation enable the formation of a large variety of compounds.
- LO2: Distinguish saturated from unsaturated carbon compounds by identifying the type of C-C bonds present.
- LO3: Identify common hydrocarbons and classify them as alkanes, alkenes, alkynes or aromatic (benzene).
- LO4: Describe the typical physical properties of hydrocarbons and relate them to molecular structure.
- LO5: Summarize the significance of Wöhler's synthesis of urea for organic chemistry.

3. Prerequisite Knowledge

- Understanding of valence electrons and covalent bonding.
- Familiarity with single, double and triple covalent bonds.
- Ability to read and write simple molecular formulas.

4. Key Terms

Term	Definition
Tetravalency	The ability of carbon to form four covalent bonds with other atoms.
Catenation	The property of carbon atoms to bond to each other, forming chains, branches, or rings.
Saturated compound	A carbon compound that contains only single C-C bonds.
Unsaturated compound	A carbon compound that contains one or more double or triple C-C bonds.
Hydrocarbon	A compound composed solely of carbon and hydrogen atoms.
Alkane	A saturated hydrocarbon containing only single C-C bonds (e.g., methane, ethane).
Alkene	An unsaturated hydrocarbon containing at least one C=C double bond (e.g., ethene).
Alkyne	An unsaturated hydrocarbon containing at least one C≡C triple bond (e.g., ethyne).
Benzene	An aromatic hydrocarbon with the formula C ₆ H ₆ , featuring a ring of six carbon atoms with alternating double bonds.

5. Key Points

- Carbon forms four covalent bonds and can link to other carbon atoms, creating countless structures.
- In 1828 Friedrich Wöhler made urea from ammonium cyanate, disproving the need for a 'vital force' in organic synthesis.
- Hydrocarbons contain only carbon and hydrogen and are grouped as alkanes, alkenes, alkynes or aromatic compounds.
- Saturated compounds have only single C-C bonds; unsaturated compounds contain double or triple bonds.
- Hydrocarbons have low melting/boiling points and are poor conductors because they are held together by weak intermolecular forces.

6. Real-World Applications

Alkanes as Everyday Fuels

Alkanes such as propane and butane are used for cooking, heating and as fuel for vehicles because they are stable, release a lot of energy on combustion, and have low boiling points that make them easy to store as liquids or gases.

Think about: Why are saturated hydrocarbons like propane suitable for use as fuels?

Suggested time: 3 minutes

Carbon Catenation and Plastic Polymers

The ability of carbon atoms to form long chains (catenation) underlies the creation of polymeric materials such as polyethylene, which are made from repeating alkane units.

Think about: How does carbon's capacity to link with itself lead to the formation of everyday plastics?

Suggested time: 4 minutes

7. Core Notes

Carbon's Tetravalency and Catenation

Carbon has an atomic number of 6, giving it four valence electrons. Because it can share all four electrons, each carbon atom can form up to four covalent bonds. This tetravalency lets carbon bond with many different elements and, importantly, with other carbon atoms. The ability of carbon atoms to join together (catenation) produces straight-chain, branched-chain and cyclic structures, providing the basis for the huge variety of organic compounds.

- Four single bonds per carbon atom
- Can link to other carbons → chains, branches, rings

Historical Breakthrough - Wöhler's Synthesis of Urea

Early chemists believed that organic compounds could only be produced by living organisms because a special 'vital force' was required. Friedrich Wöhler challenged this idea by converting the inorganic compound ammonium cyanate into the organic compound urea in 1828. This experiment showed that organic molecules can be created in the laboratory without any living system, opening the field of modern organic chemistry.

Classification of Hydrocarbons

All compounds made solely of carbon and hydrogen are called hydrocarbons. They are classified by the type of carbon-carbon bonds present:

- Alkanes - only single bonds (saturated). Example: methane (CH_4), ethane (C_2H_6).
- Alkenes - at least one double bond (unsaturated).
- Alkynes - at least one triple bond (unsaturated).
- Aromatic hydrocarbons - contain a ring with alternating double bonds, the classic example being benzene (C_6H_6).

Saturated vs. Unsaturated Carbon Compounds

A saturated hydrocarbon (alkane) has the maximum number of hydrogen atoms attached to each carbon because all C-C bonds are single. Unsaturated hydrocarbons possess one or more multiple bonds:

- Alkenes - contain $\text{C}=\text{C}$ double bonds.
- Alkynes - contain $\text{C}\equiv\text{C}$ triple bonds.

These multiple bonds reduce the number of attached hydrogens and give the molecules different chemical reactivity and physical properties.

Physical Properties of Carbon Compounds

Compared with ionic compounds, most carbon-hydrogen compounds melt and boil at relatively low temperatures. The forces between their molecules are weak van der Waals attractions, not ionic bonds, so they do not produce ions in solution. Consequently, hydrocarbons are generally non-conductors of electricity. These properties stem from the molecular (covalent) nature of the bonds rather than a lattice of ions.

8. Activities

Molecule-Building Workshop (12 min)

Using molecular model kits, students construct methane, ethane, ethene, ethyne and benzene. They label each bond type (single, double, triple, aromatic) and discuss how carbon's tetravalency enables each structure.

Historical Debate: Vital Force vs. Laboratory Synthesis (10 min)

In small groups, students read a short excerpt on the 'vital force' theory and Wöhler's urea experiment. They then debate which view is more convincing and write a brief paragraph summarizing the significance of the 1828 synthesis.

9. Questions

- Q1. What does the term 'catenation' refer to in the context of carbon chemistry?
- Q2. How can you tell if a carbon compound is saturated or unsaturated?
- Q3. Provide the molecular formula and classification (alkane, alkene, alkyne) for methane, ethane, ethene and ethyne.
- Q4. Why do most hydrocarbons have relatively low melting and boiling points compared to ionic compounds?
- Q5. What was the significance of Friedrich Wöhler's synthesis of urea in 1828?

